$q_2 = 4\pi r_2^2 \sigma_2$. Substituting these values into equation (1.112), we get

$$\sigma_1 r_1 = \sigma_2 r_2 \qquad (1.113)$$

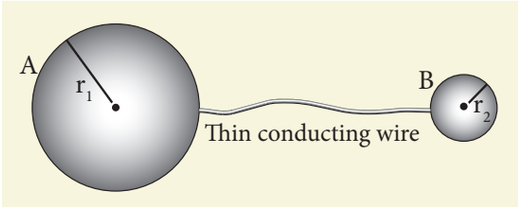
from which we conclude that

$$\sigma r = \text{constant} \qquad (1.114)$$

Thus the surface charge density σ is inversely proportional to the radius of the sphere. For a smaller radius, the charge density will be larger and vice versa.

EXAMPLE 1.23

Two conducting spheres of radius r_1 = 8 cm and r_2 = 2 cm are separated by a distance much larger than 8 cm and are connected by a thin conducting wire as shown in the figure. A total charge of Q = +100 nC is placed on one of the spheres. After a fraction of a second, the charge Q is redistributed and both the spheres attain electrostatic equilibrium.



(a) Calculate the charge and surface charge density on each sphere.

(b) Calculate the potential at the surface of each sphere.

Solution

(a) The electrostatic potential on the surface of the sphere A is $V_A = \frac{1}{4\pi\varepsilon_\circ}\frac{q_1}{r_1}$

The electrostatic potential on the surface of the sphere A is $V_B = \frac{1}{4\pi\varepsilon_\circ}\frac{q_2}{r_2}$

Since $V_A = V_B$. We have

$$\frac{q_1}{r_1} = \frac{q_2}{r_2} \Rightarrow \quad q_1 = \left(\frac{r_1}{r_2}\right) q_2$$

But from the conservation of total charge, Q = $q_1 + q_2$, we get $q_1 = Q - q_2$. By substituting this in the above equation,

$$Q - q_2 = \left(\frac{r_1}{r_2}\right) q_2$$

so that $$q_2 = Q\left(\frac{r_2}{r_1 + r_2}\right)$$

Therefore,

$$q_2 = 100 \times 10^{-9} \times \left(\frac{2}{10}\right) = 20nC$$

and $q_1 = Q - q_2 = 80\text{nC}$

The electric charge density for sphere A is

$$\sigma_1 = \frac{q_1}{4\pi r_1^2}$$

The electric charge density for sphere B is

$$\sigma_2 = \frac{q_2}{4\pi r_2^2}$$

Therefore,

$$\sigma_1 = \frac{80 \times 10^{-9}}{4 \times 64 \times 10^{-4}} = 0.99 \times 10^{-6} C m^{-2}$$

and

$$\sigma_2 = \frac{20 \times 10^{-9}}{4\pi \times 4 \times 10^{-4}} = 3.9 \times 10^{-6} C m^{-2}$$

Note that the surface charge density is greater on the smaller sphere compared to the larger sphere ($\sigma_2 \approx 4\sigma_1$) which confirms the result $\frac{\sigma_1}{\sigma_2} = \frac{r_2}{r_1}$.

The potential on both spheres is the same. So we can calculate the potential on any one of the spheres.

$$V_A = \frac{1}{4\pi\varepsilon_\circ}\frac{q_1}{r_1} = \frac{9 \times 10^9 \times 80 \times 10^{-9}}{8 \times 10^{-2}} = 9kV$$